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```
clc; clearvars; %close all; format compact
```

Constants

```
N = 30; % arbitray value
h = 2/N;
alpha = .9; % arbitray value
k1 = h^2 / 2*alpha;
k2 = 7*k1; % proves that k can exceed h^2/2*alpha
nu1 = alpha*k1/h^2;
nu2 = alpha*k2/h^2;
```

Define x axis

```
i = 0:N;
x = -1 + i*h;
```

Define t axis

```
j = 0:4*N;
t1 = j*k1;
t2 = j*k2;
```

Define original heat distribution

The heat distribution is chosen to be 1 at all intermediate locations, and 0 at either end

```
u = ones(numel(j), numel(i)); % making fake heat distribution
u(1, 1) = 0; % initial condition
u(1, end) = 0; % initial condition

% Make two different u vectors in order to draw a comparison to the
% previous problem
u1 = u;
u2 = u;

% Create the A matrix for both values of k
A1 = spdiags([-nu1, 1+2*nu1, -nu1], -1:1, N+1, N+1);
```

```
A1 = full(A1);
A2 = spdiags([-nu2, 1+2*nu2, -nu2], -1:1, N+1, N+1);
A2 = full(A2);

% go through all the rows and do the forward euler for both sets
for a = 1:numel(j)-1
    u_cur = u1(a, :);
    u_next = A1 \ u_cur';
    u1(a+1, :) = u_next';
end

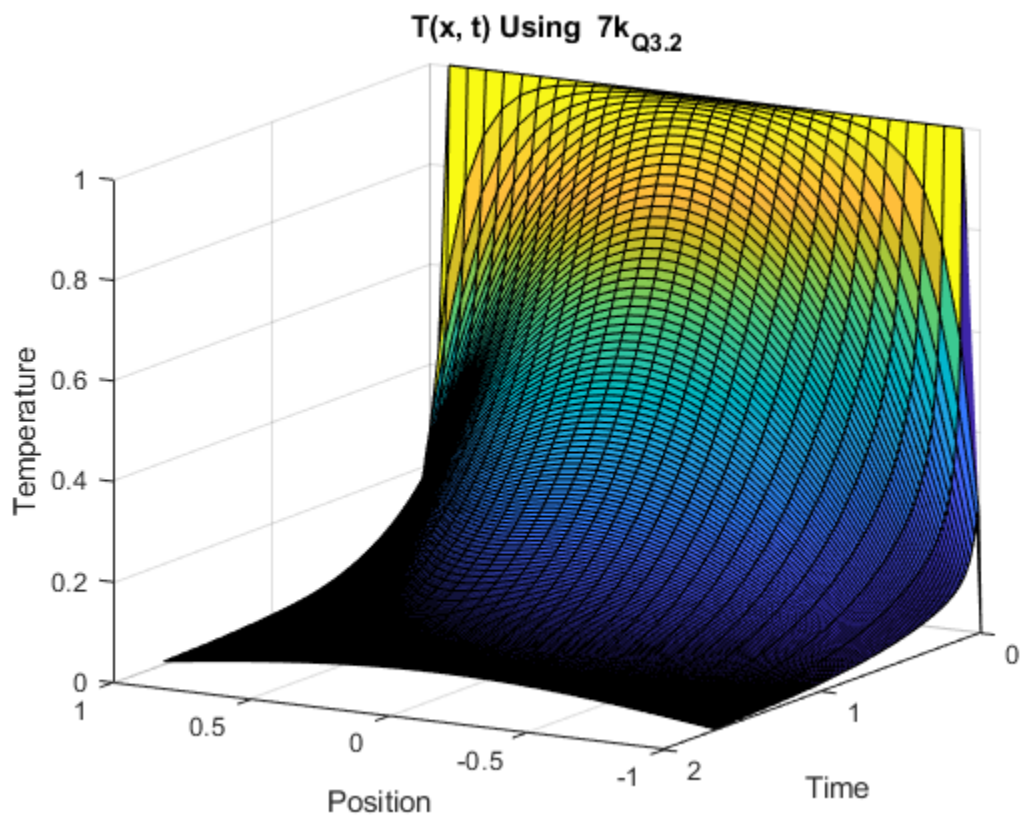
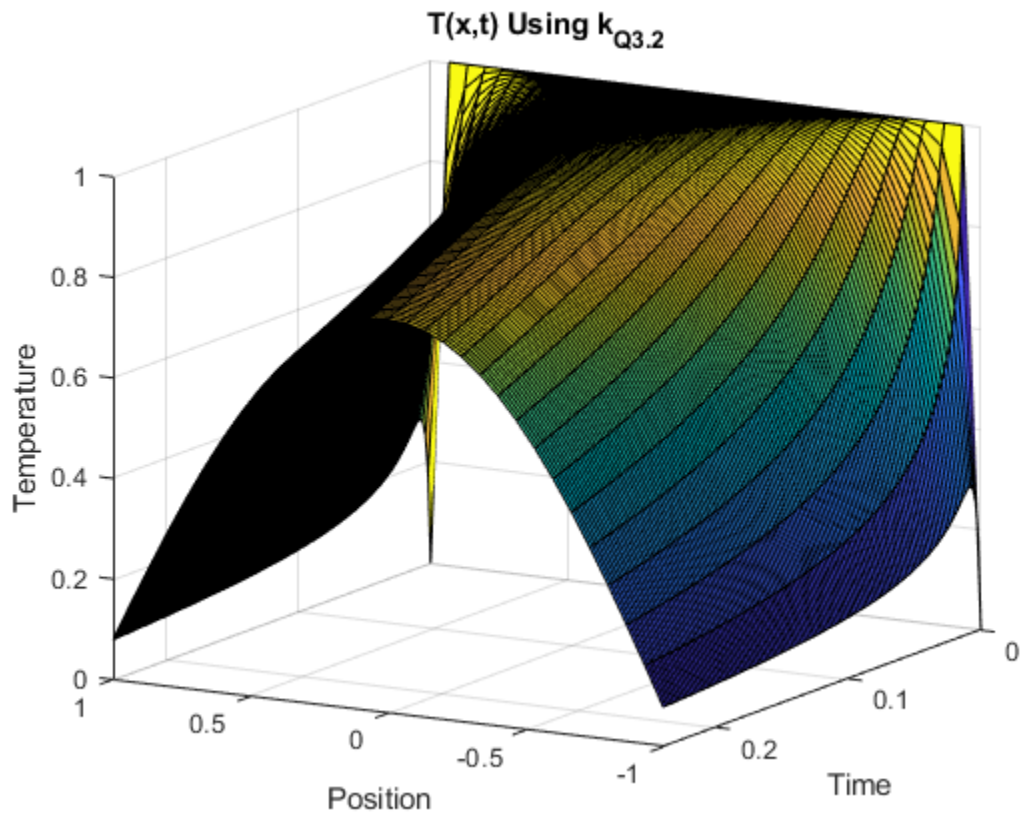
for a = 1:numel(j)-1
    u_cur = u2(a, :);
    u_next = A2 \ u_cur';
    u2(a+1, :) = u_next';
end
```

Plotting

```
[X1, T1] = meshgrid(x, t1);
[X2, T2] = meshgrid(x, t2);

figure(1)
surf(X1, T1, u1);
xlabel('Position')
ylabel('Time')
zlabel('Temperature')
view(210, 15)
title("T(x,t) Using k_{Q3.2}");

figure(2)
surf(X1, T2, u2);
xlabel('Position')
ylabel('Time')
zlabel('Temperature')
view(210, 15)
title("T(x, t) Using 7k_{Q3.2}");
```



Discussion

type `q3discussion.txt`

In this problem, q3.2 is redone but with a different method to solve for successive temperature distributions. In Figure 1, we see that the temperature distribution is the same as it was in q2. This is because in this plot, the same value of k (temporal stepsize) was used. The method used in q3.2 was limited to a maximum value of k . This means that a larger number of calculations must be completed in order to represent a large timespan.

Figure 2 shows why this new method is useful. This figure was generated using the same number of temporal steps, but is able to simulate a much longer span of time. This is useful as it lets us predict the temperature distribution as long as is needed without sacrificing compute time. In a more complex problem with more dimensions, this issue of more compute time would only compound upon itself, making solving time exponentially longer.

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